

Power Electronics Degree Examination: Mepco R2013 Regulation

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get me all the numerical or problem questions from part A

Based on the provided excerpts, the numerical or problem-based questions found in **Part A** (typically questions 1–10) are listed below:

- **Thyristor Current Calculation (AC Voltage Controller):** A single-phase full-wave AC voltage controller feeds power to a resistance of $10\ \Omega$. The source voltage is 200 VRMS. For a firing angle of 90° , determine the rms value of the thyristor current in amperes.
- **Average Thyristor Current (AC Voltage Regulator):** Find the average thyristor current for a single-phase AC voltage regulator which feeds power to a resistive load of $8\ \Omega$ from a 230 V, 50 Hz source.
- **Harmonic Component Analysis (3 ϕ Rectifier):** A six-pulse thyristor rectifier bridge is connected to a balanced 50 Hz, 3 ϕ source. Assuming the DC output current is constant, identify the frequency of the lowest harmonic component in the AC source line current.
- **PWM Harmonic Elimination:** In single-pulse modulation of PWM inverters, determine the pulse width required to eliminate the third harmonic.
- **DC Chopper Ripple Analysis:** In DC choppers, determine at which duty cycle (α) the per-unit ripple is maximum.
- **AC Output Power Control Condition:** For a single-phase AC voltage controller with an RL load, identify the relationship between the firing angle (α), load phase angle (ϕ), and conduction angle (γ) required to control the AC output power.

organize the questions like what, how , which .like that .. from part A

Based on the provided exam papers, the questions from **Part A** have been organized below according to their starting words or nature (What, How, Which, List, Compare, Draw, etc.):

"What" and "What is meant by" Questions

- **What are the advantages** of three-phase controlled rectifiers over single-phase controlled rectifiers?
- **What is the control range** of firing angle in an AC voltage controller with an RL load?
- **What is** cycloconverters?
- **What is** a dual converter?
- **What are the disadvantages** of a frequency modulation scheme over a pulse width modulation scheme?
- **What are the disadvantages** of harmonics in the inverter?
- **What are the two types** of resonant switching used in choppers?
- **What is the merit and demerit** of a voltage source inverter when compared to a current source inverter?
- **What is** integral cycle control in an AC voltage controller?
- **What is** a frequency changer?
- **What is meant by** cycloconverter?
- **What will happen** if the gate current is removed from an SCR when the anode current is 45 mA (given latching is 50 mA and holding is 40 mA)?

"How" Questions

- Mention **how the output frequency is varied** in the case of an inverter by PWM control.
- Justify **how the output frequency is varied** in the case of an inverter by PWM control.

"Which" Questions

- **Which** of the statements regarding sequence control of AC voltage controllers are true (improving power factor, reducing harmonics, or wider control)?
- From a given cycloconverter circuit diagram, **which** pairs conduct in one cycle together?

"Compare," "Contrast," and "Differentiate" Questions

- **Compare** semi-converter and full converter.
- **Compare** zero current switching and zero voltage switching of converters.

- **Distinguish** between on-line and off-line UPS.
- **Distinguish** between AC and DC filters.
- **Compare** AC voltage controller and cycloconverter.
- **Compare** half-wave and full-wave AC voltage controllers.
- **Compare the features** between integral control and phase angle control used in AC voltage controllers.
- **Compare the features** between circulating current mode and non-circulating current mode in dual converters.
- **Differentiate between** constant frequency and variable frequency control strategies used in DC choppers.
- **Compare and contrast the feature** between ON-OFF control and phase angle control.
- **Compare and contrast the features** between power MOSFET and power BJT.
- **Compare** voltage source inverter and current source inverter.
- **Compare and contrast the features** between two-pulse half-controlled and full-controlled converter circuits.
- **Differentiate** integral cycle and phase angle control.

"Define" Questions

- **Define** inverter gain.
- **Define** latching and holding current in SCR.
- **Define** displacement factor and power factor in phase-controlled converters.
- **Define** Duty ratio.
- **Define** cycloconverter and point out its applications.
- **Define** modulation index.
- **Define** integral cycle control.

"List," "Mention," "State," and "Enumerate" Questions

- **List** the various turn-on methods of SCR.
- **List four applications** of current source inverters.
- **Mention the advantages** of matrix converters.
- **List the methods** for varying the duty cycle of a DC chopper.
- **Mention any two purposes** of over-modulation.
- **List any two applications** of cycloconverters.
- **List the different types** of PWM techniques.
- **Mention some applications** of frequency changers.
- **Mention the purpose** of snubber circuits used in power electronics.
- **List the disadvantages** of variable frequency systems in DC choppers.

- **Mention the advantages** of PWM inverters.
- **List out the effects** of source impedance in controlled rectifiers.
- **Mention various control methods** used in DC choppers.
- **List out various performance parameters** of an inverter.
- **Enumerate some of the applications** of AC voltage controllers.
- **List out the types of gating signals** applicable for single-phase AC voltage controllers with R-L loads.
- **Mention the reason why** self-commutated switches are mostly preferred in chopper circuits.
- **State the significance** of the drift region in power semiconductor devices.

"Draw" and "Sketch" Questions

- **Draw** the VI characteristics of TRIAC.
- **Draw** the control characteristics of a step-down DC chopper.
- **Sketch** the power circuit of a buck-boost converter.
- **Draw** the circuit diagram of a series resonant inverter.
- **Draw** the circuit diagram of two-stage sequence control.
- **Draw** the circuit diagram of a Buck converter.
- **Draw the configuration** of a step-down chopper and give the expression for duty cycle.

Problem-Based and Reasoning Questions

- **Predict the effect** of source inductance on the output voltage of a rectifier.
- **Why** are feedback diodes required in inverters feeding inductive loads?
- **Give the merits** of IGBT over MOSFET.
- **Give the function** of a freewheeling diode in a single-phase semi-converter.
- **Give the limitations** of integral cycle control in AC to AC converters.
- **Give the expression for** RMS output voltage of a single-phase full-wave AC voltage controller with RL load for discontinuous load current.
- **Find/Compute/Determine:** Various numerical Part A questions require calculating duty cycle, RMS output voltage, or ripple frequency for specific circuits like step-up/step-down choppers or 3-phase rectifiers.

get me all the numericals or problem based questions from part A , Part B and Part C .. organize them

Based on the provided exam papers, the numerical and problem-based questions are organized below by section.

Part A (Short Numerical & Reasoning Questions)

- **Thyristor Current Calculation:** For a single-phase full-wave AC voltage controller feeding a $10\ \Omega$ resistance with a source voltage of 200 VRMS and a firing angle of 90° , determine the RMS value of the thyristor current.
- **Average Current Calculation:** Find the average thyristor current for a single-phase AC voltage regulator feeding power to a resistive load of $8\ \Omega$ from a 230 V, 50 Hz source.
- **Harmonic Component Analysis:** Identify the frequency of the lowest harmonic component in the AC source line current for a six-pulse thyristor rectifier bridge connected to a balanced 50 Hz, 3 ϕ source.
- **PWM Pulse Width Determination:** Determine the pulse width required to eliminate the third harmonic in single-pulse modulation of PWM inverters.
- **Ripple Analysis:** In DC choppers, identify at which duty cycle (α) the per-unit ripple is maximum.
- **Ripple Frequency Determination:** For a single-phase full bridge converter fed from 230 V, 50 Hz supplying a highly inductive load, calculate the frequency of the voltage ripple on the DC side.
- **RMS Voltage Estimation:** For a single-phase full-wave AC voltage controller with an input voltage of 230 V (RMS) and a firing angle of 90° , estimate the approximately resulting RMS output voltage.
- **VSI Fundamental Component:** For a single-phase full-bridge VSI fed from a 300 V battery with a pulse duration of 120° , calculate the RMS value of the fundamental component of the output voltage.
- **Gate Removing Scenario:** Predict what happens if the gate current is removed from an SCR when the anode current is 45 mA, given a latching current of 50 mA and holding current of 40 mA.

Part B (Descriptive Numerical & Design Problems)

Rectifiers and Phase Controlled Converters

- **1 ϕ Fully Controlled Converter:** A converter is supplied from 230 V, 50 Hz with a load of $R = 10\ \Omega$ and a large inductance. For a triggering angle of 30° , estimate average load voltage, average output current, and input power factor.
- **1 ϕ Semi-Converter Performance:** A semi-converter is connected to a 120 V, 60 Hz supply. Assuming continuous load current, calculate the input current harmonic factor (HF), displacement factor (DF), and input power factor.
- **3 ϕ Half-Wave Converter:** Operated from a star-connected 208 V, 60 Hz supply with a $10\ \Omega$ load. Calculate the delay angle required to obtain an

average output voltage of 50% of the maximum possible voltage, along with rectification efficiency and transformer utilization factor (TUF).

- **3 ϕ Fully Controlled Converter:** Supplied from a 400 V, 50 Hz system with a constant DC load current of 100 A and firing angle of 60°. Calculate the average DC voltage and input power.
- **RLE Load Problems:** A single-phase full converter feeds an RLE load ($R = 6\ \Omega$, $L = 6\text{ mH}$, $E = 60\text{ V}$) from 230 V, 50 Hz. Find the average load current for a firing angle of 50° and calculate the new current if one SCR becomes open-circuited.

Choppers and DC-DC Converters

- **Step-Up Chopper:** A chopper with a pulse width of 150 μs operates on a 220 V DC supply. Compute the load voltage if the blocking period is 40 μs .
- **Buck Converter Design:** Design the filter components (L and C) for a buck converter with an input of 12 V, output of 8 V, peak-to-peak ripple voltage of 20 mV, and peak-to-peak ripple current of 0.8 A at 25 kHz.
- **Boost Converter:** Given an input of 6 V and an average output of 18 V with a load current of 0.4 A, calculate the required duty cycle and design parameters.
- **Buck-Boost Regulator:** Given $V_s = 12\text{ V}$, duty cycle = 0.25, and switching frequency = 25 kHz, calculate the average load voltage and efficiency.
- **Time Ratio Control:** For a chopper operating at 2 kHz on a 220 V supply with a load voltage of 170 V, compute the conduction and blocking periods of the thyristor in each cycle.

Inverters and AC-AC Converters

- **1 ϕ Full-Wave AC Voltage Controller:** Given a 10 Ω resistive load and 230 V, 50 Hz input, determine the load RMS voltage, power observed by the load, and average thyristor current for a delay angle of 90°.
 - **Integral Cycle Control:** A controller supplies a 20 Ω lamp from a 220 V, 50 Hz source. If thyristors are ON for 30 cycles and OFF for 70 cycles, calculate the RMS output voltage and input power factor.
 - **Cycloconverter Analysis:** For an input of 120 V (RMS), 60 Hz and a load of $R = 5\ \Omega$, $L = 40\text{ mH}$, determine the RMS output voltage and input power factor for an output frequency of 20 Hz.
 - **Heater Power Calculation:** A 230 V, 1 kW heater is fed through a one-pulse converter. Find the power absorbed by the heater for firing angles of 45° and 90°.
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Part C (Application and Design Cases)

(Note: While many sources do not explicitly label a "Part C," the following high-level construction/analysis problems are typically found in the final section of the exam papers.)

- **Bidirectional Converter Construction:** Construct and explain a suitable bidirectional power converter to feed a 160 V, 50 Hz lamp load from a 230 V, 50 Hz source.
- **Frequency Converter Design:** Construct a power converter to obtain 100 Hz and 25 Hz from a 50 Hz input source and explain using circuit diagrams and waveforms.
- **Star Connected Inverter Analysis:** Derive the actual and fundamental RMS voltage and power for a 3 ϕ inverter with 180° conduction mode for a star-connected resistive load.

The following questions from **Part B** (Question Nos. 11, 12, 13, 14) are frequently repeated across the provided exam papers:

Question 11: Power Semiconductor Devices

- **Thyristor (SCR) Turn-on and Two-Transistor Analogy:** Explain the turn-on mechanism or operation of a thyristor based on the **two-transistor model/analogy** and derive the expression for anode current,,,,,,,,.
- **IGBT Characteristics:** Describe the construction, principle of operation, and both **static (I-V) and switching characteristics** of the Insulated Gate Bipolar Transistor (IGBT) with necessary diagrams,,,,,,,,.
- **Power MOSFET Characteristics:** Explain the switching, steady-state, or transient characteristics of enhancement-type **Power MOSFETs**,,,,,.
- **Series Connection of SCRs:** Explain the series connection of SCRs and derive the equations for **static and dynamic equalization/sharing circuits**,,,,,.
- **Power Diode and Protection:** Explain the structure and operation of power diodes, including their **reverse recovery characteristics**,,,,, and describe typical **snubber circuits** used for thyristor protection,.,

Question 12: Phase Controlled Converters

- **Single-Phase Fully Controlled Converter:** Explain the operation and performance of a **single-phase full converter** with R or RL/RLE loads, including derivations for average output voltage and relevant waveforms,,,,.
- **Single-Phase Half-Controlled (Semi-Converter):** Discuss the operation of a **single-phase semi-converter** with RL/RLE loads, often involving the

use of a freewheeling diode and calculations for average output voltage and power factor,,,,,,.

- **Three-Phase Converters:** Explain the operation of **three-phase full converters** or three-pulse (half-wave) converters with RL/resistive loads and derive the output voltage equations,,,,.
- **Single-Phase One-Pulse Converter:** Derive expressions for the average output voltage and analyze the performance of a **one-pulse (half-wave) controlled converter** with resistive loads,,,,.

Question 13: DC-DC Converters (Choppers)

- **Step-Up Chopper:** Explain the operation of a **step-up DC chopper** (Boost converter) and derive the expression for its average output voltage,,,,.
- **Buck Converter Design:** Describe the working principle of a **Buck (step-down) converter** and design its filter components (L and C) based on given ripple constraints,,,,,,,,.
- **Buck-Boost Converter:** Explain the operation of a **Buck-Boost regulator** (step up-down) and analyze its performance parameters through derivation and numerical problems,,,,,,,,.
- **Chopper Control Strategies:** Discuss different control strategies used in choppers, such as **Time Ratio Control (TRC)** and variable frequency control,,.
- **Four-Quadrant Operation:** Explain the concept and working of a **four-quadrant DC chopper** with neat circuit diagrams,,.

Question 14: Inverters

- **Three-Phase Inverter (180° Mode):** Explain the operation of a **three-phase bridge inverter** in **180° conduction mode** with necessary phase and line voltage waveforms for star-connected loads,,,,,,,,.
- **Three-Phase Inverter (120° Mode):** Describe the working of a **three-phase bridge inverter** in **120° conduction mode** with relevant waveforms,,.
- **PWM Techniques:** Discuss various **Pulse Width Modulation (PWM)** techniques used for voltage control in inverters, such as single-pulse, multiple-pulse, and sinusoidal PWM,,,,,,,,.
- **Single-Phase Full Bridge Inverter:** Explain the working of a **single-phase full bridge voltage source inverter (VSI)** with R and RL loads,,.
- **UPS Systems:** Explain the working principle of **online and offline Uninterrupted Power Supply (UPS)** systems,,.
- **Current Source Inverters (CSI):** Describe the operation of a **single-phase current source inverter**, often featuring capacitor commutation and resistive loads,,.

similarly repeated questions in 15 q.no

Based on the provided exam papers, the questions for **Question No. 15** (which typically covers **AC-AC Converters**, including AC Voltage Controllers and Cycloconverters) are categorized into the following repeated themes:

1. AC Voltage Controllers (Regulators)

- **Integral Cycle Control (ON-OFF Control):** Explain the principle of integral cycle control or ON-OFF control with necessary circuit diagrams, waveforms, and derivations for RMS output voltage.
- **Numerical Problems (ON/OFF Cycles):** A single-phase AC voltage controller with a resistive load is given specific ON and OFF cycles (e.g., 6 cycles ON, 4 cycles OFF; or 30 cycles ON, 70 cycles OFF). Calculate the RMS output voltage, input power factor, and thyristor currents.
- **Sequence Control:** Discuss or analyze the construction and operation of **two-stage sequence control** of AC voltage controllers (often using lamp loads) with circuit diagrams and waveforms.
- **Unidirectional vs. Bidirectional Controllers:** Explain the operation and analysis of unidirectional and bidirectional AC voltage controllers for R and RL loads, including circuit diagrams and waveforms.
- **Analysis of RL Loads:** Illustrate the circuit diagram and operation of a single-phase full-wave AC voltage controller feeding an **RL load**, often requiring derivations for RMS load current and power factor.
- **Specific Applications:** Explain the operation of a converter used for **speed control of fans** in domestic applications using AC supply.

2. Cycloconverters (Frequency Changers)

- **Single-Phase Step-Down Cycloconverters:** Explain the working principle and operation of single-phase to single-phase **step-down cycloconverters** using **bridge-type** or **midpoint configurations** with neat circuit diagrams and waveforms.
- **Frequency Conversion Design (Construction):** Construct a suitable power converter using SCRs (often eight) to obtain specific output frequencies from a given input frequency (e.g., 60 Hz to 30 Hz; 100 Hz to 50 Hz; or obtaining 100 Hz and 25 Hz from a 50 Hz source).
- **Step-Up Cycloconverters:** Describe the concept of **step-up cycloconverters** and identify suitable power electronic switches for this operation.
- **Phase Control in Frequency Changers:** Explain the operation of frequency changers when phase control techniques are employed to obtain variable AC output voltage and frequency.

3. Specialized Converters & Triggering

- **Zero Voltage Triggering:** Explain the operation of **zero voltage triggering** based AC voltage controllers, typically assuming a heater load.
- **Bidirectional Design Case:** Construct a bidirectional power converter to feed a specific lamp load (e.g., 102 V, 50 Hz) from a higher input source (e.g., 230 V, 50 Hz).
- **Derivations:** Derive the expression for the **RMS output voltage** for various control techniques like phase control or ON-OFF control in single-phase regulators.

What are the common Part C design questions?

Explain the 180 degree conduction mode in detail

Show me the repeated questions for Question 11